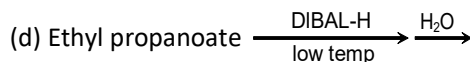
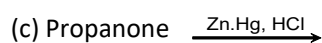
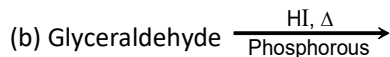
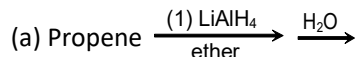


SCQ (Single Correct Type) :

1. In which reactions major product is propane ?



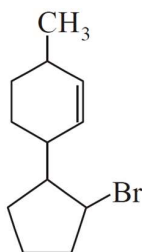
(A) a & c

(B) b & d

(C) a & d

(D) b & c

2. IUPAC name of the compound is



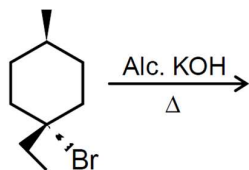
(A) 3 - (2 - bromo cyclopentyl) - 6-methyl - cyclohex - 1 - ene

(B) 3 - methyl - 6 - (2 - bromo cyclopentyl) - cyclohex - 1 - ene

(C) 1 - methyl - 4 - (2 - bromo cyclopentyl) - cyclohex - 2 - ene

(D) 4 - methyl - 1 - (2 - bromo cyclopentyl) - cyclohex - 2 - ene

3. For the following reaction, the correct option is:



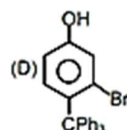
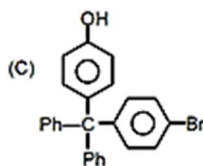
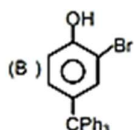
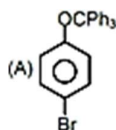
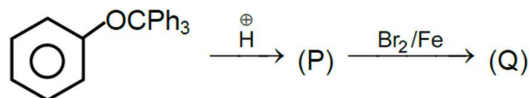
(A) Three organic products are obtained.

(B) Four organic products are obtained in which two are chiral.

(C) Two enantiomeric pairs are one obtained and both pairs are constitutional isomers with each other

(D) Only two organic products are obtained

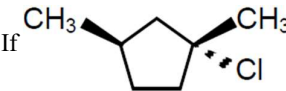
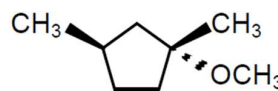
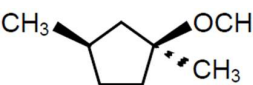
4. The major product (Q) obtained in the following reaction is :



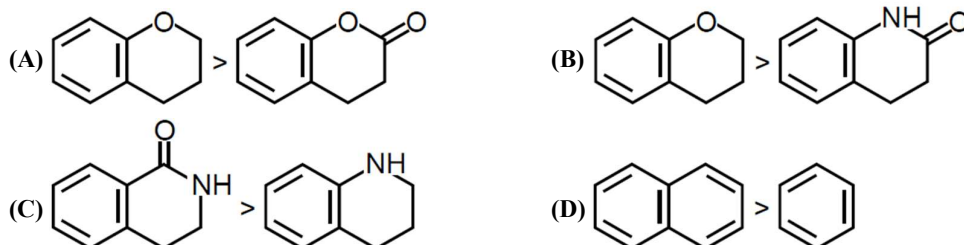
5. $\xrightarrow[\text{H}_2\text{O}]{\text{Br}_2}$ (P) major product
- (A) (B) (C) (D)
6. $\xrightarrow{\text{H}^+/\Delta}$ Product is
- (A) (B) (C) (D)
7. Which of the following will produce chiral molecule(s) after reaction is completed.
- (A) $\xrightarrow{\text{NaOH/CaO}, \Delta}$
- (B) $\xrightarrow{\text{PCC}}$
- (C) $\text{PhCHO} \xrightarrow[\text{H}_2\text{SO}_4]{(\text{i}) \text{NaCN}}$
- (D) $\xrightarrow[\text{(2) NH}_4\text{Cl}]{\text{(1) CH}_3\text{MgBr}}$

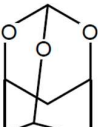
MCQ (Multiple Correct Type) :

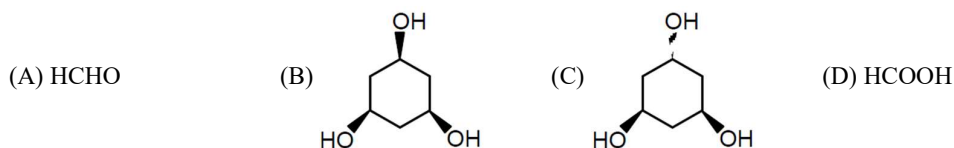
8. For Cyclooctatetraene following is correct
- (A) There are two types of C – C bond.
- (B) Structure is non planar and resonance is not observed in molecule.
- (C) Extensive resonance is found with in the molecule and all bonds are of same type.
- (D) Its heat of hydrogenation is equal to that of 4π bonds hydrogenated.
9. Which of the following show intramolecular hydrogen bonding?
- (A) salicylaldehyde (B) o-nitrotoluene
- (C) p-nitrophenol (D) hydrogenphthalate ion $\left(\text{C}_6\text{H}_4(\text{COOH})(\text{COO}^-) \right)$
10. Correct statement is/are regarding carboxylic acids having molecular formula $(\text{C}_8\text{H}_8\text{O}_2)$ containing aromatic ring.
- (A) The total number of carboxylic acids (with one benzene ring) is 4.
- (B) Out the three Toluic acids, the ortho isomer is the strongest acid.
- (C) All these Toluic acids are weaker acids than benzoic acid.
- (D) All these are weaker acids than H_2CO_3 .

11. If  is reacted with methyl alcohol (CH_3OH) both  and  are obtained. The correct statement/s about the above reaction is/are.
- (A) It is an example of acid-base reaction.
 (B) Its rate equation is independent of concentration of CH_3OH .
 (C) In above reaction, the pH will decrease.
 (D) Alkene as a major product is obtained at room temperature.

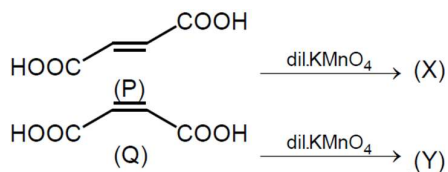
12. Which of the following is/are correct order of reactivity towards electrophilic substitution reaction with Br_2/Fe .



13.  $\xrightarrow{\text{H}_3\text{O}^+}$ The product obtained in following reaction is/are.

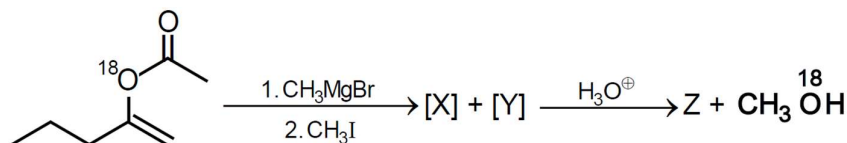


14. Which of the following statement(s) is/are correct for above pair of reactions.



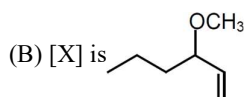
- (A) Reactions are stereospecific and stereoselective
 (B) Above reaction are an example of electrophilic addition reactions
 (C) (Y) is mesotartaric acid while (X) is racemic mixture
 (D) The most stable conformation of (Y) is Gauche form.

15. Observe the following reaction, and answer as per given reaction only



If Y gives yellow precipitate with NaOI then which of the following statement(s) is/are correct.

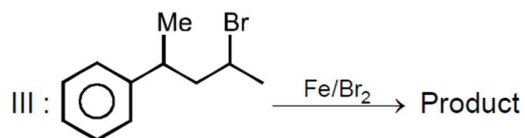
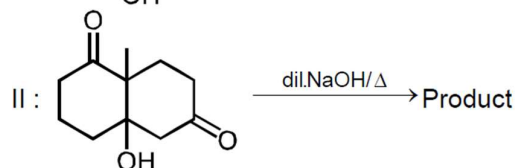
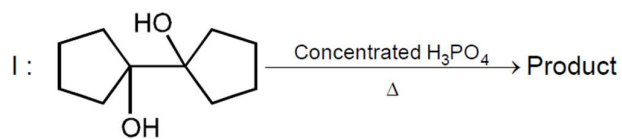
(A) [X] is an enol ether



(C) [Z] gives butanoic acid on reaction with bleaching power followed by acidification.

(D) [Z] is ketone.

16. Which of the following is/are correct for given reactions ?



(A) Reaction I has a spirocyclic compound as a product.

(B) Reaction II has a carbanion intermediate.

(C) Reaction I & II both involve dehydration.

(D) Reaction I, II, III all are elimination reactions.

Match the column

Match the reactions given in List-I with the result of the respective reactions with appropriate options given in List-II.

	LIST-I		LIST-II
(I)	Number of structural isomeric alcohols of $C_4H_{10}O$	(P)	3
(II)	Number of structural isomeric esters of $C_4H_8O_2$	(Q)	5
(III)	Number of structural isomers of C_6H_{14}	(R)	6
(IV)	Number of structural isomers of $C_2BrClFI$	(S)	4
		(T)	7
		(U)	8

17. Which of the following has the correct combination considering List-I and List-II ?

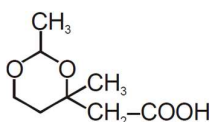
(A) (I - S) (B) (II - S) (C) Both (A) & (D) None of these

18. The total number of isomers of $C_2BrClFI$?

(A) (P) (B) (Q) (C) (R) (D) (S)

(Paragraph) :

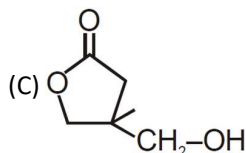
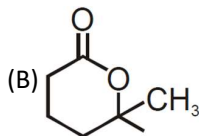
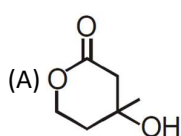
A research scholar synthesised the compound A in the laboratory



He wanted to study this compound in acidic medium and kept for some time with heating. After an hour, when isolated the compound, to this surprise, he got the mixture of two compounds B, C. He studied their properties and compound them with A

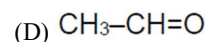
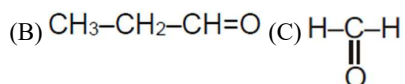
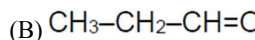
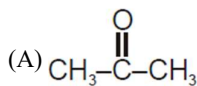
A	B	C	
Reaction with NaHCO_3	$\uparrow \text{CO}_2$	No reaction	No reaction
Blue litmus	red	No	No
NaOI	No	No	Yellow ppt
$[\text{Ag}(\text{NH}_3)_2]^+$	No	No	Silver mirror
[Hint : $\text{A} \xrightarrow{\text{H}_3\text{O}^+/\Delta} \text{B} + \text{C}$]			

19. Compound B is :



(D) None

20. Compound C is :



21. B is formed due to :

(A) Cleavage of C - O bond followed by esterification between $\square\text{-COOH}$ & -OH present.

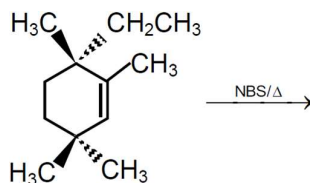
(B) Decarboxylation of $\square\text{COOH}$ group

(C) Oxidation of $\square\text{-CH}_2\text{COOH}$ into $\square\text{COOH}$

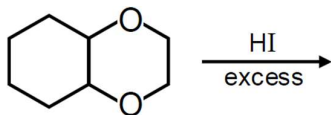
(D) None

(Numerical Value type) :

22. How many different products (including stereoisomers) are obtained in the following reaction?



23. Calculate number of moles of HI involved in the following reaction.



Answer key

1.	(D)	2.	(A)	3.	(C)	4.	(B)	5.	(C)
6.	(C)	7.	(C)	8.	(ABD)	9.	(AB)	10.	(AB)
11.	(BC)	12.	(ABD)	13.	(BD)	14.	(ABC)	15.	(ACD)
16.	(ABC)	17.	(C)	18.	(C)	19.	(A)	20.	(D)
21.	(A)	22.	3	23.	6				